

Nathan Chan

Biochemistry graduate with hands-on protein expression, purification, and analytical characterization experience. Seeking a research position in peptide and protein science.

EDUCATION

University of Washington, Seattle — B.S. Biochemistry

September 2022 - June 2026

GPA 3.8/ 3.7 STEM

Relevant Coursework: Biochemistry I-III, Organic Chemistry/Lab I-III, Immunology, Biochemical Thermodynamics & Kinetics I-II, Medicinal Chemistry: Science of Drugs (PK/PD, CYP450 metabolism, SAR, biologics)

North Seattle College, Seattle — EMT

June 2024- August 2024

EMT training in pre-hospital medicine, building strong understanding of human physiology, rapid decision-making, and teamwork-based patient care.

RESEARCH EXPERIENCE

University of Washington — Biochemistry Lab, 4.0

- Designed the W123Y point substitution in β -glucosidase B (BglB) and modeled the predicted structural effect in Foldit, targeting W123 for its role in substrate positioning and active site packing.
- Carried out the full experimental workflow: site-directed mutagenesis, Sanger sequence verification, bacterial transformation, recombinant expression in *E. coli*, affinity chromatography purification, and SDS-PAGE purity confirmation.
- Characterized the purified variant by spectrophotometric kinetic assay. The mutant produced a near-flat Michaelis-Menten curve with $V_{\max}$ and K_m approaching zero, yielding very low k_{cat} and catalytic efficiency.
- Concluded that tyrosine substitution introduces polarity disrupting residue-substrate interaction and active site packing, consistent with the Foldit prediction. Full report: chan602.com/gallery.html#capstone

WORK EXPERIENCE

SBP, Momentum Climbing Sodo — Instruction, Operations

October 2022 - August 2025, August 2025 - PRESENT

Instructed beginner and intermediate climbers in technique and safety; managed facility operations and wall maintenance. Provided front desk customer service; recognized for top performance in membership sales.

Healthcare Alternative Spring Break (HCASB) — Team Lead

January 2026 - March 2026

Led 6 undergraduates on a week-long clinical shadowing program at Forks Community Hospital. Managed logistics, scheduling, and community outreach; facilitated healthcare career presentations to local students.

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TECHNICAL SKILLS

Western Blot/ SDS-PAGE

Enzyme Activity Assays

Plasmid Vector/Bacteria Transformation

Sanger Sequencing

Affinity Chromatography

Site-Directed Mutagenesis

Foldit, ChimeraX

IR, NMR, Mass Spec., GC/MS

Javascript/React.js,
HTML/CSS

Git/[Github](https://github.com)

Microsoft/Google Suites

AWARDS & CERTIFICATIONS

UW Department of Chemistry and Biochemistry Graduation Speaker, 2026

Four Year Dean's List (2022-2026)

National Registry of Emergency Medical Technicians (NREMT) (Exp 3/31/27)

Prudential Volunteer Service Award (227hrs)

SAT (1590)

LANGUAGES

English, Cantonese